Climatologies at High resolution for the Earth Land Surface Areas

High resolution global standardized drought indices

Release Date: 28. 01. 2025

Document version: 1.1

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CHELSA: File Specification

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data should be cited as:

Scientific publication:

Chen, L., Brun, P., Buri, P., Fatichi, S., Gessler, A., McCarthy, M. J., Pellicciotti, F., Stocker, B., Karger, D. N. (2025). Global increase in the occurrence and impact of multiyear droughts. ***Science*** https://doi.org/10.1126/science.ado4245

Data citation:

Chen, L., Brun, P., Buri, P., Fatichi, S., Gessler, A., McCarthy, M. J., Pellicciotti, F., Stocker, B., Karger, D. N. (2024). High resolution global standardized drought indices. ***EnviDat***. https://www.doi.org/10.16904/envidat.530

Revision history

Version	Date	Changes
1.0	31.07.2024	Initial document
1.1	28.01.2025	Added 10km data and qkndvi

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1. Introduction

The dataset consists of the standardized precipitation (SPI), the standardized precipitation evapotranspiration index (SPEI) index at 30 arcsec. resolution, and identified multiyear droughts with a 1/24° (~5km) and 1/12° (~10km) horizontal resolution, as well as anomalies in kernel normalized difference vegetation index (QKNDVI) at 1/12° (~10km) horizontal resolution.

Data for precipitation rates (pr) and potential evapotranspiration (pet) are taken from CHELSA V2.1 (https://doi.org/10.16904/envidat.228.v2.1). Both data sets are monthly time series from 1980 to 2018 at 30 arcsec. Horizontal resolution with global coverage that were aggregated to 1/24° (~5km) spatial resolution and have the unit of kg m-2 month-1. SPEI is a more reliable measure of drought than SPI as it additionally considers the effect of potential evapotranspiration. Potential evapotranspiration (pet) is the amount of water per area per time that can evaporate at the soil surface or be transpired by plants in absence of water supply limitations. SPEI effectively estimates for a given location how the climatic water balance, i.e., the difference between precipitation rate (pr) and potential evapotranspiration (pet), relates to the long-term mean. Here, pet was calculated with the Penman-Montheith approach, assuming surface conductance of a reference crop of 12 cm height following the definition of the Food and Agriculture Organization of the United Nations (FAO), originating from CHELSA-BIOCLIM+ database (https://www.doi.org/10.16904/envidat.332). To obtain global time series of SPEI, in each pixel the frequency distribution of water balance estimates (pr-pet) is approximated with a log-logistic probability distribution function. We calculate monthly SPEI using a 12-month memory and approximate the parameters of the log-logistic probability distribution based on the time series from 1980-2018. Such a 12-month time window is suitable to capture long-term drought events that are long enough to impose substantial impacts on agriculture, hydrology, and ecosystems while ones at shorter time scales would be more appropriate to detect short-term drought. For each month, SPEI is calculated considering the respective water-balance value and the values of the eleven preceding months. Here we use SPEI-12 in December to represent the overall water-balance condition of the respective year. The resulting SPEI data has a horizontal resolution of 1/24° (~5km). The calculation of SPI and SPEI is done in R with the package SPEI.

Multiyear droughts were classified into drought and non-drought conditions using a threshold value. A widely-used categorization scheme for SPEI considers the values from 0 to -0.99 as mild drought, from -1 to -2 as moderate to extreme drought. Here, we target (multiyear droughts) MYDs with sustained, severe, and intense dry conditions. To capture these conditions, we set the SPEI to -1.1 for thresholding drought conditions.

Then, we apply a mathematical morphology approach to identify spatiotemporally contiguous MYD events globally during the period 1980-2018. To this end, the binary (drought/non-drought) grids were first stacked (in chronological order) and the resulting stack is interpreted as a three-dimensional cuboid, with longitude, latitude, and time as dimensions, and voxels of 1/24° × 1/24° × 12 months resolution. In addition, only drought voxels that are identified as being part of events that last at least two consecutive years (with direct neighbors in the time dimension) are retained. As we focus on multiyear events, we remove drought voxels from this cuboid that are not adjacent to the neighboring drought voxels which are defined as the other 26 voxels around the target voxel in a cubic consisting overall 27 voxels (three voxels in each dimension, 3 × 3 × 3 = 27). Moreover, we exclude voxels in regions that are classified as "hot desert" and "cold desert" by the Köppen-Geiger climate classification system, as co-occurring changes in vegetation greenness are hard to detect in these sparsely vegetated regions and SPEI anomalies are difficult to interpret in deserts. Then, the spatiotemporal neighborhood of each drought voxel is screened for adjacent drought voxels, and contiguous voxels are assigned to the same drought ID. This approach corresponds to the view that a multiyear drought is an event that propagates through space and time.

The start and end of a MYD event are given by the years of its first and last voxel, respectively. Notably, a pixel that initially belongs to a MYD in a given year may no longer belong to the MYD in subsequent years before it merges back into the same MYD event again. Therefore, multiple seemingly spatiotemporally independent drought events can be part of a single MYD event, granted that they concatenate spatially at some time point before the overall end of a MYD event. The MYDs identified from this method consist of at least two voxels in the dimension of time, representing the minimum two-year drought condition.

The identification of contiguous events is carried out using the conncomp_3D function of the neuroim2 package in R.

B. Buchsbaum, *Neuroim2: Data Structures and Handling for Neuroimaging Data* (2022; https://github.com/bbuchsbaum/neuroim2)vol. R package version 0.1.0.

2. Format and File Organization

All files are provided as cloud optimized georeferenced tiff files (GeoTIFF). GeoTIFF is a public domain metadata standard which allows georeferencing information to be embedded within a TIFF file. Additional information included in the file are: map projection, coordinate systems, ellipsoids, datums, and fill values.

All GeoTIFF files are saved as integer with a compression = deflate, predictor = 2, and an internal scale and offset in case of continuous variables. Scale and offset are internally stored in the GeoTIFF files and are supported by GDAL Version 2.3 or higher.

In some cases, problems have been reported using older versions of GDAL or ArcGIS. In this case the scale and offset must be set manually for the variables.

GeoTIFF can be viewed using standard GIS software such as:
SAGA GIS – (free) http://www.saga-gis.org/
ArcGIS - https://www.arcgis.com/
QGIS - (free) www.qgis.org
DIVA – GIS - (free) http://www.diva-gis.org/
GRASS – GIS - (free) https://grass.osgeo.org/

2.1 Dimensions

All CHELSA-BIOCLIM+ files contain a variable that define the dimensions of longitude and latitude. The time variable is usually encoded in the filename.

Table 3.1. Dimension Variables

Name	Description	type	Attribute
longitude	Longitude	double	degrees_east
latitude	Latitude	double	degrees_north
variable	variable	int	variable

2.2 Variables

All variables of CHELSA are time-averaged and contain either monthly, annual means, or long term normals (climatologies), but not mixtures of these. Instantaneous parameters are not provided. Monthly time-averaged are files usually based on means of synoptic hours. Monthly files represent averages for the calendar months, accounting for leap years. For monthly means, each file contains a single month.

3. Grid Structure

All global CHELSA products are in a geographic coordinate system referenced to the WGS 84 horizontal datum, with the horizontal coordinates expressed in decimal degrees. The CHELSA layer extents (minimum and maximum latitude and longitude) are a result of the coordinate system inherited from the 1-arc-second GMTED2010 data which itself inherited the grid extent from the 1-arc-second SRTM data.

The CHELSA-drought-indices data has different kinds of grid extents:

Grid extent spi:

Attribute		**value**
Resolution		0.0083333333
West extent (minimum X-coordinate, longitude):		−180.0001388888
South extent (minimum Y-coordinate, latitude)		−90.0001388888
East extent (maximum X-coordinate, longitude)		179.9998611111
North extent (maximum Y-coordinate, latitude)		83.9998611111
Rows		20,800
Columns		43,200

Note that because of the pixel center referencing of the input GMTED2010 data the full extent of each CHELSA grid as defined by the outside edges of the pixels differs from an integer value of latitude or longitude by 0.000138888888 degree (or 1/2 arc-second). Users of products based on the legacy GTOPO30 product should note that the coordinate referencing of CHELSA (and GMTED2010) and GTOPO30 are not the same. In GTOPO30, the integer lines of latitude and longitude fall directly on the edges of a 30-arc-second pixel. Thus, when overlaying CHELSA with products based on GTOPO30 a slight shift of 1/2 arc-second will be observed between the edges of corresponding 30-arc-second pixels.

Grid extent spei:

Attribute		**value**
Resolution		0.0083333333
West extent (minimum X-coordinate, longitude):		−180.0001388888
South extent (minimum Y-coordinate, latitude)		-56.9376390171
East extent (maximum X-coordinate, longitude)		179.9998611111
North extent (maximum Y-coordinate, latitude)		82.2540270928
Rows		16,704
Columns		43,200

Grid extent mymd:

Attribute		**value**
Resolution		0.0416666665
West extent (minimum X-coordinate, longitude):		-179.9793055557
South extent (minimum Y-coordinate, latitude)		-56.9293056837
East extent (maximum X-coordinate, longitude)		179.9790263378
North extent (maximum Y-coordinate, latitude)		82.2373604263
Rows		3341
Columns		8640

Grid extent mymd10 and qkndvi:

Attribute		**value**
Resolution		0.083333
West extent (minimum X-coordinate, longitude):		-179.9584722225
South extent (minimum Y-coordinate, latitude)		-56.9501390172
East extent (maximum X-coordinate, longitude)		179.9581930045
North extent (maximum Y-coordinate, latitude)		82.2165270928
Rows		1671
Columns		4320

4. File Naming Conventions

The filename of each CHELSA data product follows a similar structure including the respective model used, the variable short name, the respective time variables, and the accumulation (or mean) period in the following basic format:

CHELSA_[short_name]_[timeperiod]_[Version].tif

5. Variable Names

Table 7.1 gives the variable short names, longnames, units, scale, offsets, and explanations. Scale and offset are internally stored in the GeoTIFF files and are supported by GDAL Version 2.3 or higher. In some cases, problems have been reported using older versions of GDAL or ArcGIS. In this case the scale and offset has to be set manually for the variables. This can be done by first multiplying the raster values with the 'scale' value and then adding the 'offset' value.

7.1. monthly means

shortname	longname	unit	scale	offset	explanation
spi	Standardized precipitation index	-	0.01	-300	The standardized precipitation index
spei	Standardized precipitation evapotranspiration index	-	0.01	-300	The standardized precipitation evapotranspiration index

7.2. Annual

shortname	longname	unit	scale	offset	explanation
mymd	Multiyear meteorological drought	ID	1	0	ID of the multiyear meteorological drought at 1/24° horizontal resolution. The ID is a unique identifier that characterizing a pixel belonging to a specific MYMD with the same ID.
mymd10	Multiyear meteorological drought	ID	1	0	ID of the multiyear meteorological drought at 1/12° horizontal resolution. The ID is a unique identifier that characterizing a pixel belonging to a specific MYMD10 with the same ID.
qkndvi	Kernel normalized difference vegetation index anomaly	-	1	0	Anomaly of the kernel normalized difference vegetation index.